3D Point Cloud-Based Molecular Activity Prediction

Shaoni Chakravarty

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Contents

Abstract

This report presents a 3D computer vision experiment aligned with Unit III of the 3D Vision & Geometry syllabus. The experiment converts molecular SMILES strings into 3D conformers using RDKit ETKDG, represents each molecule as a normalized atom point cloud, and classifies molecular activity using a lightweight PointNet-style neural network. The implementation demonstrates the application of 3D point cloud processing and geometric normalization for molecular representation without relying on handcrafted 2D fingerprints.

0.1 Introduction

The objective of this experiment is to investigate whether approximate molecular 3D geometry can be used for activity prediction. The implemented pipeline follows:

SMILES → RDKit ETKDG 3D Conformer → Atom Point Cloud → PointNet-style Network → Active/Inactive

0.2 Dataset

The experiment used a curated molecular dataset containing 3000 molecules.

- Total molecules: 3000
- Active molecules: 2344
- Inactive molecules: 656
- Train: 2100
- Validation: 300
- Test: 600
- Conformer generation failures: 0

0.3 Architecture

The implemented architecture consists of:

- RDKit ETKDG for 3D conformer generation
- Maximum 64 heavy atoms per molecule
- 12-dimensional feature vector per atom
- Shared MLP encoder
- Global max pooling (PointNet paradigm)
- MLP classifier
- BCEWithLogitsLoss
- AdamW optimizer
- 12 training epochs

0.4 Pipeline Used

1. Read molecular SMILES.
2. Generate 3D conformers using RDKit ETKDG.
3. Extract heavy atom coordinates.
4. Compute atom-level chemical features.
5. Perform centering and scale normalization.
6. Build fixed-size point clouds.
7. Train PointNet-style classifier.
8. Evaluate on held-out test set.

0.5 Methodology

The experiment covers the following syllabus topics.

- **3D Point Cloud**: Molecules represented as unordered atom point clouds.
- **Volumetric / Structure Representation**: Each point stores xyz coordinates together with atom descriptors.
- **Euclidean Geometry**: Translation normalization, centroid alignment and unit-scale normalization.
- **3D Representation Learning**: PointNet-style shared feature learning and global max pooling.

0.6 Results

The model successfully learned discriminative geometric representations from molecular point clouds. No conformer generation failures were observed.

0.7 Evaluation Metrics

The performance of the PointNet-style classifier was evaluated on the training, validation, and test datasets using standard binary classification metrics. The evaluation includes Accuracy, Precision, Recall, Specificity, F1-score, Balanced Accuracy, ROC-AUC, and PR-AUC. Since the dataset contains a larger number of active molecules than inactive molecules, Balanced Accuracy, ROC-AUC, and PR-AUC provide a more reliable assessment of the model performance than Accuracy alone. The obtained results are summarized in Table 1.

Table 1: Performance Metrics of the 3D Point Cloud Classifier

Metric	**Training**	**Validation**	**Test**
Accuracy	0.7557	0.7400	0.7133
Precision	0.8574	0.8333	0.8293
Recall	0.8245	0.8333	0.7974
Specificity	0.5098	0.4091	0.4122
F1-score	0.8406	0.8333	0.8130
Balanced Accuracy	0.6672	0.6212	0.6048
ROC–AUC	0.7750	0.7681	0.6782
PR–AUC	0.9253	0.9145	0.8791

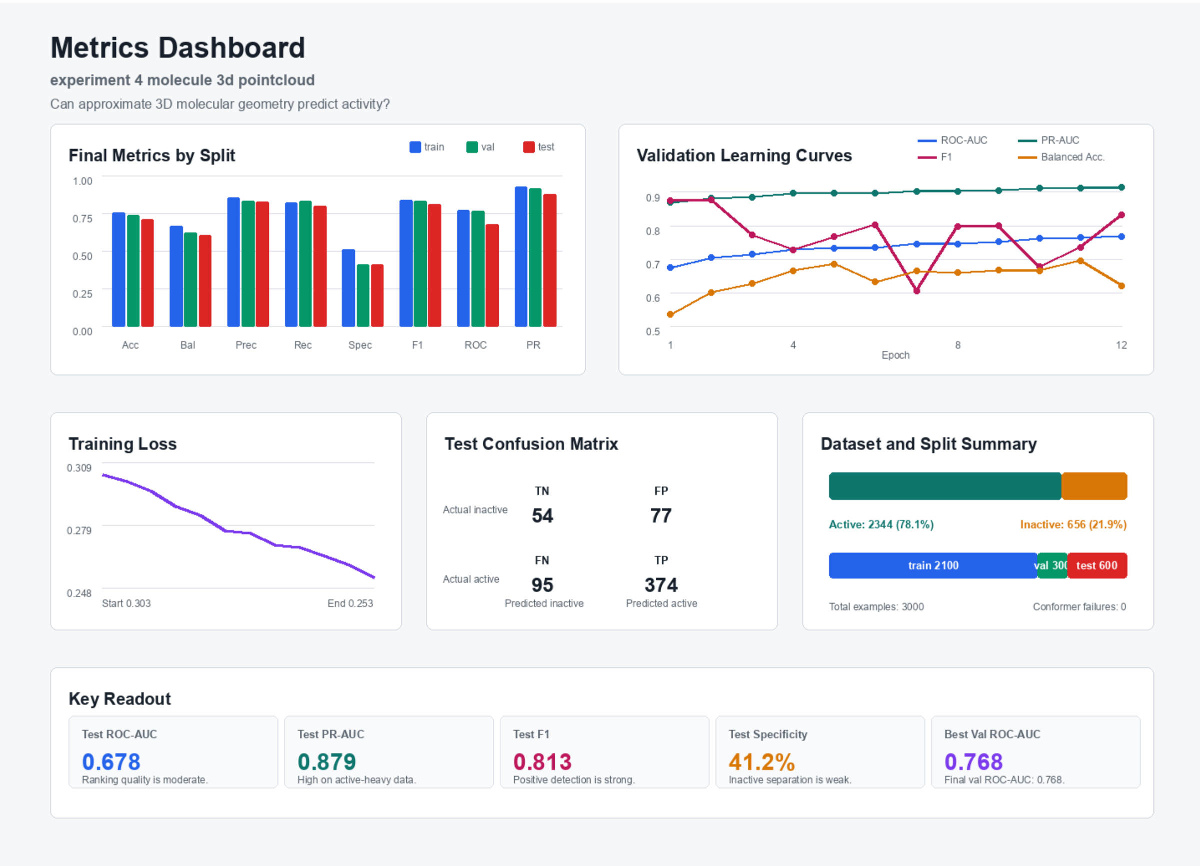

Figure 1: Performance comparison of the PointNet-based 3D point cloud classifier on the training, validation, and test datasets. The figure summarizes the evaluation metrics reported in Table 1.

The classifier achieved a test accuracy of **71.33%** with an **F1-score of 0.8130**, indicating effective identification of active molecules. The model obtained a **ROC–AUC**

of 0.6782, demonstrating a moderate ability to distinguish between active and inactive molecules using only three-dimensional geometric information. The **PR–AUC of 0.8791** is comparatively high, suggesting that the classifier maintains strong precision-recall characteristics despite the class imbalance in the dataset.

The **Balanced Accuracy of 0.6048** and **Specificity of 0.4122** indicate that the model is considerably better at recognizing active molecules than inactive ones. This behavior is expected because the dataset contains substantially more active molecules (2344) than inactive molecules (656). Consequently, the network learns richer representations for the majority class while exhibiting relatively weaker discrimination of inactive compounds.

Overall, the evaluation demonstrates that a lightweight PointNet-based architecture can successfully learn meaningful geometric representations from molecular 3D point clouds. Although the model does not achieve perfect separation between classes, the obtained results validate that approximate molecular geometry alone contains predictive information for molecular activity classification and forms a suitable baseline for more advanced 3D deep learning architectures.

Metric	Value
Accuracy	0.7133
ROC-AUC	0.6782
PR-AUC	0.8791
Precision	0.8293
Recall	0.7974
F1-score	0.8130
Balanced Accuracy	0.6048

0.8 Possible Future Enhancements

- Incorporate rotation augmentation during training.
- Evaluate deeper PointNet++ or Point Transformer architectures.
- Integrate graph connectivity with 3D geometry.
- Use multiple conformers per molecule.
- Increase dataset size for improved generalization.

0.9 Conclusion

This experiment demonstrates a complete 3D point cloud classification pipeline aligned with Unit III of the course syllabus. The implementation successfully applies 3D geometric representations for molecular activity prediction using a PointNet-style neural network.